

When Does the Second View Help? Auditing Cross-View Fusion in Urban Vision-Language Models

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Paper ID #*****

Abstract. Pairing overhead satellite and ground-level street imagery is widely assumed to give vision-language models a richer view of a place than either alone. We test when this assumption holds. We build OELLM, a deterministic, OpenStreetMap-grounded benchmark that pairs a marked satellite tile with road-aligned street views across 40 cities, covering urban-attribute tasks about one location and cross-view tasks whose answer depends on the correspondence between the two views. We pose every question three ways—satellite only, street only, and both—and read off *fusion synergy*, the gain of both views over the better single one. On the seven urban-attribute tasks, where we find each single view already carries the answer, a fine-tuned 9B VLM gains only +0.5 points from adding the second view; looking question by question, the second view in fact flips more answers from right to wrong than from wrong to right, so it is mildly harmful rather than merely redundant. The cross-view tasks, built so the answer lives between the views, behave oppositely and confirm the same ablation does detect fusion when a task requires it. The urban pattern replicates across model families and scales, stronger prompting, inference-time view pruning, and training-time view count. Whether a second view helps is task-shaped, not pervasive.

Keywords: Geosensing · Vision-language models · Multi-view fusion

1 Introduction

More perspectives are assumed to be better: a satellite tile shows layout and connectivity, a street-level view shows facades, surfaces, and signage, and pairing them should give a vision-language model a richer picture of a place than either alone. A growing line of work collects, aligns, and jointly models satellite and ground views for urban understanding on exactly this premise. But “should” is rarely tested directly—existing benchmarks report one accuracy over a mix of tasks, some answerable from a single view and some requiring both, so a high number can mean genuine fusion or simply that one view sufficed. We ask the question the aggregate hides: *when does the second view actually help?*

We answer it with a controlled audit. We build OELLM, a deterministic, OpenStreetMap-grounded benchmark that pairs a marked satellite tile with

road-aligned street-level views across 40 cities. It spans two kinds of task: urban-attribute questions about one location (land use, building height, transit access), and cross-view questions whose answer depends on the correspondence between the two views. Our instrument is an image ablation—we pose each question with the satellite tile alone (*sat*), the street view alone (*sv*), or both (*full*)—from which we read *fusion synergy*, the gain of both views over the better single one, $\text{full} - \max(\text{sat}, \text{sv})$. We evaluate at the minimum input that makes the comparison fair—one satellite tile plus one forward street view—so the second view adds exactly one image, and apply the audit to a fine-tuned 9B VLM and a panel of raw open VLMs. Figure 1 previews the setup and the headline effect.

The result is a clean dissociation. On the seven urban-attribute tasks the second view adds only +0.5 points over the better single view on average; looking item by item, it does not merely fail to help but is mildly *harmful*, flipping more answers from right to wrong than the reverse (Sec. 4.4). The cross-view tasks, by contrast, are built so the answer lives between the views, and there the same model does gain from holding both—confirming the audit can register fusion when the task demands it, rather than being blind to it. The urban pattern is robust: it reproduces across raw VLMs of different families and scales, survives a 16k-token reasoning budget and a multi-turn self-check, holds when inference-time pruning keeps a single street view, and persists when the model is trained with all four street views rather than one.

This is a *task-dependent* observation at commonly served satellite resolutions, not a claim that overhead imagery is uninformative or that fusion fails in general—the cross-view tasks show the same models gaining when fusion is required.

Our contributions are:

1. **A task-dependence result for cross-view fusion.** On seven urban-attribute tasks a fine-tuned 9B VLM gains essentially nothing (+0.5 points) from a second view over the better single one, while on cross-view tasks in the same benchmark it gains substantially—a within-model dissociation showing that pairing helps only when the task structurally requires it. The question is how current models *use* two informative views, not whether one view alone could suffice.
2. **An item-level interference analysis** showing the second view is not merely redundant on these tasks but, item by item, mildly harmful: it over-returns more correct answers than it rescues, so a model that simply picked the better single view per question would outscore the two-view model (Sec. 4.4).
3. **OELLM, a deterministic paired-view benchmark** of 12 OSM-grounded task families over 40 cities, whose image ablation and two task groups make the single-view-sufficient versus fusion-required distinction measurable.
4. **A simple decision rule for practitioners:** before paying to collect a second imagery modality, measure its marginal value with the ablation we describe (does full beat the better single view, per item?). We show the verdict here is stable across model families and scales, inference-time view pruning, and training-time view count.

2 Related Work

Remote-sensing vision-language models are mostly overhead-only. A wave of instruction-tuned VLMs target remote sensing, including GeoChat [5], SkySenseGPT [14], LHRS-Bot and LHRS-Bot-Nova [8,16], EarthGPT [25], and the multi-sensor generalist EarthDial [18]. These are overhead-only: each consumes a single aerial or satellite image and none ingests ground-level imagery (Tab. 2). Where “multi-modal” appears here it typically refers to multiple *overhead* sensors—optical, SAR, infrared—rather than to pairing satellite with street-level views. The dominant suites follow suit: RSVQA [13], VRSBench [7], and GeoChat’s own benchmark pose questions over a single overhead image, and none isolates the marginal value of a second, ground-level view. Architecturally these build on LLaVA-style designs [10,11] with CLIP or SigLIP encoders [17,24]; we use four as zero-shot baselines and discuss their training-data provenance in Sec. 4.1.

Multi-view urban benchmarks. A smaller set of benchmarks pairs overhead and ground views and notices the surface phenomenon we study, but stops short of explaining it. UrBench [27] reports that multi-image and single-image performance differ little on coordinate-aligned street/satellite pairs, but attributes the gap to question *comprehension* and measures only aggregate accuracy, leaving open whether the second view is unused, redundant, or actively harmful. CityLens [12], concurrent with our work, finds street view alone competitive with both views—but on a three-task socioeconomic *regression* target, with a single model and no item-level or cross-view analysis. Neither isolates *where in input-mode space* the effect lives. We contribute the missing instrument: an image ablation that exposes per-item interference rather than only an aggregate, applied across a task taxonomy whose two groups let the same instrument register fusion when the task requires it. Our result is not “a second view rarely helps”—already suspected—but a quantified *effect* (it removes more correct answers than it adds) and a *boundary* (fusion is decisive exactly when the answer lives between views). UrbanLLaVA [2] trains over satellite, street-view, OSM, and trajectory inputs but poses its attribute tasks from satellite alone; CityCube [22] targets cross-view spatial reasoning rather than attribute recognition.

Cross-view geo-localization: where both views are necessary. Satellite-ground pairing is well established in cross-view geo-localization, where the goal is to match a ground photo to an overhead reference [1,23,28]. There the two views are jointly *necessary*, and methods succeed by modeling the distinct-but-linkable layout they share [26]. This is the regime our cross-view tasks occupy; complementarity that is constitutive of localization is largely absent from single-view-decidable attribute recognition.

The marginal value of a second modality. That adding a modality need not help is known beyond geosensing. Wang *et al.* [20] show the best single-

On urban-attribute tasks the second view adds ≈ 0 over the better single view; the large cross-view gains are from tasks built to require both views.

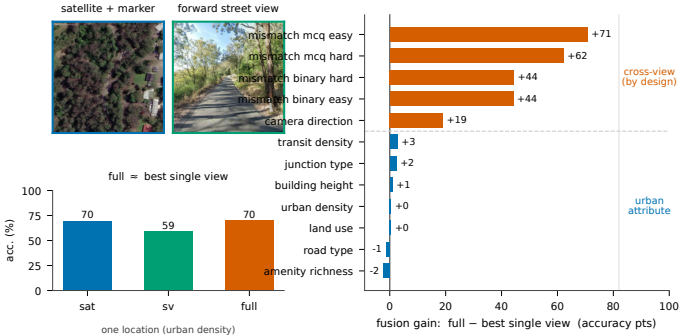


Fig. 1: Left: one held-out location (Sydney). Each benchmark location provides a marked sub-meter satellite tile and four cardinal street-level views; the dataset always supplies all four. Our primary model consumes only *two* images—the satellite tile and the *forward* street view—and the ablation is computed in that configuration: “full” here means satellite + forward street view. For this task (urban density) full barely exceeds the better single view. **Right:** per-task fusion gain (full – better single view) across all 12 evaluated tasks. On the seven urban-attribute tasks the gain is ≈ 0 (some negative); the large gains on the five cross-view tasks come from tasks constructed so the answer lives between the views, where removing one view is costly by design. Imagery is unmodified dataset content: satellite tile © Esri World Imagery; street-view panels © Google (Street View).

modal network can outperform a naively fused one, and Liang *et al.* [9] formalize multimodal interaction into redundancy, uniqueness, and synergy, showing low-synergy regimes are real; our “fusion synergy” is the behavioral analogue of their synergy term. Against this, multimodal learning can provably lower achievable risk [4], and several aerial+ground systems report recognition gains [15,21]. Most directly counter to our setting, Suel *et al.* [19] find fusing satellite and street imagery improves *regression* of income and deprivation; we engage this in Sec. 6, noting it differs in task type (regression vs. four-way recognition), labels, and model. Our finding refines rather than contradicts the “fusion helps” literature by naming a regime where it does not.

3 The OELLM Benchmark

We build **OELLM**, a deterministic, OpenStreetMap-grounded benchmark that pairs overhead and street-level imagery to make the question “*when does a second view help?*” measurable. Every answer derives from OSM tags or coordinate geometry rather than human annotation. The benchmark is limited to what OSM and geometry can certify, but every answer is verifiable and the same OSM snapshot regenerates it.

EOLLM pipeline — seven generation stages

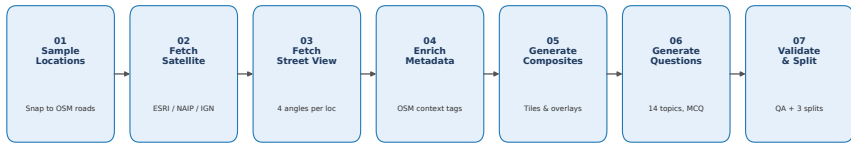


Fig. 2: The seven-stage deterministic generation pipeline. Every stage reads and enriches the same list of sample dictionaries; no human annotation is involved.

3.1 Construction

Each *location* is sampled within one of 40 cities across 32 countries on six continents and is associated with (i) one satellite tile, (ii) four street-level views at fixed road-aligned bearings (forward, backward, left, right), and (iii) OSM-derived attribute labels. A seven-stage pipeline (Fig. 2) threads each sample through location sampling, satellite-tile fetching, street-view capture, OSM enrichment, composite rendering, deterministic question generation, and validation/splitting.

Satellite imagery is sourced per location by geography, always from a sub-meter aerial orthophoto: ESRI World Imagery (~ 0.3 m/px to 0.5 m/px in cities) as the global default, NAIP (~ 0.6 m/px) in the US, and IGN (~ 0.2 m/px) in France. Across the 34,484-record corpus, 31,489 questions (91%) draw ESRI imagery, 1946 NAIP, and 1049 IGN, so the redundancy we report later is measured against uniformly sub-meter overhead imagery. The satellite tile carries a marker indicating the queried location; the marker is part of the input and is held fixed across all conditions (its effect is examined in Sec. 5). Street-level views are retrieved from Google Street View and retain Google’s original attribution; satellite imagery is © its respective provider (Esri, NAIP/USDA, IGN).

3.2 Task taxonomy

The benchmark contains 12 task families in two groups (Fig. 3): seven urban-attribute tasks and five cross-view tasks.

Urban understanding (seven tasks). `land_use`, `building_height`, `urban_density`, `road_type`, `junction_type`, `amenity_richness`, and `transit_density`. Each is a four-way multiple-choice question over a marked satellite tile and street-level imagery of one location, with the answer determined by that location’s OSM context. These are the tasks on which we ask whether a model gains from holding both views.

Cross-view geolocalisation (five tasks). `camera_direction` asks which of four arrows on the satellite tile points in the heading a single street view was captured

from; `mismatch_binary_{easy,hard}` ask whether a street-view set belongs to a satellite tile; `mismatch_mcq_{easy,hard}` ask which of four street-view composites matches the tile. *Easy* variants draw distractors from another city (where vegetation, signage language, and vehicle types often leak the answer); *hard* variants draw them from the same city, defeating those shortcuts and isolating the geometric match. These answers live in the *relationship* between views and are unanswerable from a single view by construction.

A fair test of fusion. The two groups differ in what a second view can buy, and that difference is the design principle. On the urban tasks each single view already carries the answer: the satellite-only and street-only accuracies of our trained model track each other closely (e.g. `land_use` 72.4 vs. 71.7, `building_height` 60.0 vs. 62.6; Sec. 4.2). The question is therefore not whether one view *can* suffice but whether a model *gains* from holding both—a fair test, because each view is independently sufficient. The cross-view group is the complement, included so the same instrument can register synergy when fusion is the only route to the answer.

3.3 Splits

Splitting is done at the location level: all of a location’s questions stay together, never crossing the train/validation/benchmark boundary *within a given training strategy*. The *benchmark* is carved out first and identically—ten percent of the locations in every one of the 40 cities, the same 4734 questions (all 12 tasks) regardless of strategy—and shipped in two parallel files (`public`, no answers; `with_answers`, gold-labelled). The remaining locations are then divided two ways (Tab. 1): a *per-city* split that sends a fixed fraction of each city’s locations to validation (every city appears in both train and validation), and a *seen/unseen* split that holds out eight cities entirely to probe geographic generalisation. The two strategies draw their train and validation pools from the same remainder, so they are not mutually disjoint; each is internally clean against the shared benchmark, and we train models under both. The headline model uses the seen/unseen split. Option shuffling is seeded from `sample_id` + topic so the correct letter is stable across re-runs; we keep one question per (location, task) pair and discard locations with fewer than four street-level views.

3.4 Evaluation protocol

We evaluate every model under a controlled *image ablation*: each question is posed under three input conditions—`sat` (the marked satellite tile only), `sv` (street-level view only), and `full` (both). The text of the question and options is byte-identical across conditions; only the supplied images change, so the contrast of `full` against either single view isolates the marginal value of adding the other modality. We decode greedily with a fixed seed and parse a single answer letter with a strict word-boundary matcher; the same harness, prompt, and rendering run across all models so that only the forward pass differs. The fusion

Table 1: Record counts for the released OELLM corpus (34,484 questions over 4168 locations in 40 cities). Two training strategies share one benchmark, carved out first and identically. This paper’s fusion audit uses the seven urban-attribute tasks of the benchmark ($n = 2629$); the remaining 2105 cross-view questions provide the sensitivity check of Sec. 4.3.

Split set	Train	Validation	Benchmark
per_city	23,431	6319	—
seen_unseen	23,282	8179	—
benchmark (12 tasks)	—	—	4734



Fig. 3: Two rendered tasks. **Left** (urban understanding, `land_use`): a marked satellite tile and four street views; the answer follows from one location’s OSM context. **Right** (cross-view, `camera_direction`): four satellite crops with candidate heading arrows and one street-view query; the answer lives in the relationship between the views.

audit uses the 2629 urban-attribute questions; the 2105 cross-view questions serve as a sensitivity check on the instrument. Per-task counts are uneven by design, reflecting OSM-tag availability rather than importance (`building_height`, 190; `junction_type`, 334; the other five urban tasks, 421 each).

3.5 Coverage and limitations

Coverage is uneven in two respects. Geographically the corpus is Europe-weighted (the single largest share of records), with the remaining cities spread across five other continents; absolute accuracies should be read with that skew in mind, though our headline metric is computed within each question and is unaffected by the geographic mix. Label-wise, a few attribute tasks depend on OSM tags that are sparse in some cities—`building:levels`, which underlies `building_height`, is sparse in São Paulo and Singapore—so those tasks contribute fewer questions. Full distributional statistics are provided in the accompanying dataset report.

Table 2: Off-the-shelf remote-sensing VLMs evaluated zero-shot on the overhead (satellite-only) half of the benchmark. All four are single-image and *overhead-only*; none ingests ground-level imagery. “OSM data” marks whether the model’s training corpus is derived from OpenStreetMap (relevant to the provenance caveat in Sec. 5).

Model	Base LLM	Vision encoder	Input res.	OSM data
GeoChat-7B [5]	Vicuna-1.5-7B	CLIP ViT-L/14	504	No
SkySenseGPT-7B [14]	Vicuna-1.5-7B	CLIP ViT-L/14	336	No
LHRS-Bot-Nova [8]	Llama-3-8B	SigLIP-L/14	336	Yes
EarthDial-4B-RGB [18]	Phi-3-Mini	InternViT-300M	tiled	Yes

4 Experiments

The central result is at the item level (Sec. 4.4); we build to it—setup, the average effect, a sensitivity check—then analyse it directly.

4.1 Setup

Our primary model is **Qwen3.5-9B** fine-tuned with LoRA [3] (rank 16, $\alpha = 16$, all linear layers including the vision tower, bfloat16, max image edge 512 px) on the seen/unseen training split. We train it at the *minimum-sufficient* input—a marked satellite tile plus the single *forward* street view (*satfwd*), two images per sample—so that every comparison sits on the same one-street-view regime. Fine-tuning on the task distribution is deliberate: a failure to use both views then cannot be blamed on poor task alignment. For the mismatch tasks, which structurally require comparing the full street-view set against the satellite tile, evaluation feeds all four street views even to this adapter, because the task design forces it.

We evaluate under the three image-ablation conditions of Sec. 3.4: *sat* (marked tile), *sv* (forward street view), and *full* (both). In this configuration *full* means *two* images—one satellite, one street view; we report the four-street-view training regime separately in Sec. 4.8. The quantity of interest is *fusion synergy* = *full* − max(*sat*, *sv*), the gain of both views over the better single one. To check that the finding is not model-specific we also evaluate four off-the-shelf, overhead-only remote-sensing VLMs zero-shot (Tab. 2; Sec. 4.5) and a panel of raw open VLMs under the same conditions (Sec. 4.6); all run through one harness on a single 96 GB GPU.

4.2 The second view adds little over the better single view

Does adding the second view improve average accuracy? Table 3 reports the trained model per urban task under the three conditions. *full* tracks the *better single view* closely: per-task fusion synergy ranges from −2.4 (*amenity_richness*) to +2.9 (*transit_density*) and averages +0.5 points across the seven tasks; two tasks have *negative* synergy, and street view alone already matches or exceeds

Table 3: Trained Qwen3.5-9B (satfwd) on the held-out benchmark, per urban task and input condition (accuracy, %). **Syn.** = $\text{full} - \max(\text{sat}, \text{sv})$ is fusion synergy. Synergy is small on every task and **full** stays close to the better single view; each single view alone is already well above the 25.2% chance rate.

Task	n	sat	sv	full	Syn.
land_use	421	72.4	71.7	72.9	+0.5
building_height	190	60.0	62.6	63.7	+1.1
urban_density	421	69.6	59.1	70.1	+0.5
road_type	421	72.7	76.0	74.6	-1.4
junction_type	334	71.6	66.5	74.0	+2.4
amenity_richness	421	55.3	47.7	53.0	-2.4
transit_density	421	46.3	46.1	49.2	+2.9
Mean	—	64.0	61.4	65.3	+0.5

full on road_type. Pooled, synergy is +1.1 points (95% CI $[-0.2, +2.4]$, including zero). The model is plainly reading the imagery—each single view lands far above the 25.2% chance rate of always answering ‘A’, the lowest being transit_density at 46.1% on the weaker view—so the issue is not that the pixels go unused but that the second view’s pixels carry about a single view’s worth of usable information.

A synergy near zero is only meaningful if the instrument can detect fusion when fusion is present.

4.3 The instrument is not blind to fusion (sensitivity check)

The five cross-view tasks check that the same ablation registers fusion where fusion is genuinely present. They are constructed so the answer lives in the correspondence between the two views—and, for the mismatch tasks, so the candidate street-view set that matches the tile is literally one of the options—so removing a view is, by design, costly. It is: with both views the model reaches 87–98% on the mismatch tasks (Tab. 4), far above either single view. The instrument that records a half-point change on the urban tasks is plainly not insensitive, so the urban null is a real absence of synergy. We do *not* read these magnitudes as a fusion “score”: the comparison is not image-count-matched (the mismatch tasks feed the full street-view set), and a large drop when an answer-bearing view is withheld is expected by design. The image-count-matched question—does a second *modality* help added to the first?—is the urban one (Secs. 4.2 and 4.4), where full adds exactly one street view to one satellite tile.

4.4 Item-level: the second view interferes more than it helps

A small positive average could still hide genuine fusion on a subset of items. We join the three conditions’ predictions at the item level—every question carries a stable identifier shared across conditions—and analyse the 2629 fully paired urban items directly (Fig. 4).

Table 4: Sensitivity check: the same trained model on the cross-view tasks, whose answers depend on the relationship between the views (the mismatch tasks include the matching view among the options). The large full-vs-single gap confirms the ablation is not blind to fusion; we treat it as a check on the instrument, not as an image-count-matched measurement.

Cross-view task	sat	sv	full	Syn.
camera_direction	23.0	–	42.0	+19.0
mismatch_binary_easy	47.0	53.4	97.6	+44.2
mismatch_binary_hard	48.0	44.2	92.4	+44.4
mismatch_mcq_easy	26.1	23.8	96.9	+70.8
mismatch_mcq_hard	22.6	24.9	87.2	+62.2

The second view overturns far more correct answers than it rescues. Genuine fusion is rare: full is correct while *both* single views are wrong on only 40 items (1.5%). Interference is common: full is wrong while at least one single view was right on 363 items (13.8%). The second view therefore overturns a correct answer about nine times as often as it rescues a wrong one. Equivalently—and this is the *same* statistic stated as an upper bound—an *oracle* that keeps the better of the two single-view answers per item (a router, not a fusion model, and not an achievable system, since selecting the right view uses the gold label) beats full by 12.3 points (77.5 vs. 65.2; paired bootstrap 95% CI [10.9, 13.7]; McNemar $p < 10^{-60}$). The two-view model is beaten by a single-view router on every one of the seven tasks (Fig. 4). Against the street-only view, full is significantly *worse* than parity ($p < 10^{-4}$); against the satellite-only view the win/loss split is not significant (174 vs. 145, $p = 0.12$). When the two views disagree, the combined input does not reliably preserve the correct answer.

Stronger reasoning does not remove it. Perhaps the second view is complementary and the model merely fails to combine it—in which case stronger inference-time reasoning should surface the signal. It does not: across plain decoding, a 16k-token explicit-reasoning budget, and a four-turn observe-reason-self-check-commit protocol, mean accuracy rises modestly while fusion synergy stays *negative* and street-view-alone remains the single strongest condition (evaluated on an AWQ-quantised Qwen3.5-9B so the long-context runs fit one GPU; this probe is an internal trend check, not numerically comparable to the headline, and the claim rests only on the sign of synergy, which quantisation does not flip). More deliberation raises single-view accuracy without raising synergy.

The average gain, precisely. The pooled synergy of +1.1 points (CI [−0.2, +2.4]) is statistically equivalent to zero under a ± 3 -point TOST [6]. The *harm*, by contrast, is significant: the single-view oracle’s lead has an interval well clear of zero. Whether a near-zero gain at that interference cost warrants serving a second imagery modality is the practical question we take up in Sec. 6.

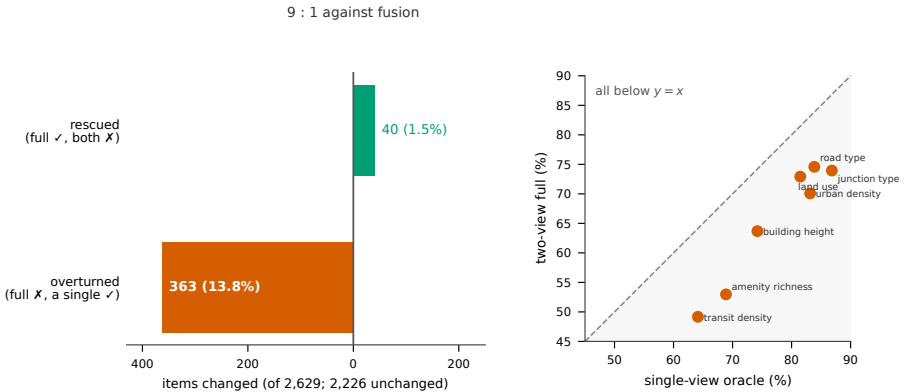


Fig. 4: Left: of the 2629 urban items, the second view changes 403—it rescues 40 (both single views missed) and overturns 363 (a single view had it right), a 9:1 imbalance against fusion. **Right:** per task, two-view accuracy (y) versus an oracle keeping the better single view per item (x ; an upper bound, not an achievable system). Every task lies below $y = x$. Trained Qwen3.5-9B (satfwd).

4.5 The finding holds zero-shot across off-the-shelf RS-VLMs

To rule out that the redundancy is an artefact of our particular model, we evaluate four off-the-shelf remote-sensing VLMs (GeoChat, SkySenseGPT, LHRs-Bot-Nova, EarthDial) zero-shot on the satellite-only half of the urban benchmark. None was trained on our task, our four-way format, or our marker overlay; they span CLIP- and SigLIP-based encoders, three LLM families, and single-crop versus tiled preprocessing. A fine-tuned model should of course beat zero-shot baselines; the comparison bears not on synergy but on the single-overhead-view floor. The best zero-shot model (EarthDial, 37.8%) trails our trained model’s satellite-only accuracy (64.0%) by 26 points. As unweighted means over the same seven urban tasks, GeoChat scores 21.0%, SkySenseGPT 27.2%, LHRs-Bot-Nova 29.5%, and EarthDial 37.8%; the latter two are trained on OSM-derived text and are an upper bound on visual skill (Sec. 5).

4.6 The regime split reproduces across families and scales

Is the dissociation specific to our fine-tuned model? The same ablation on raw, zero-shot open VLMs—Qwen3.5-4B/9B, Qwen2.5-VL-7B, and Gemma-4-12B—preserves the *shape* of the result. The raw models are weaker overall (urban full accuracy clusters near 50% across all four, versus 65% for our trained model), but the pattern is the same: a small urban fusion gain and a large cross-view fusion gain. On the raw Qwen3.5-9B, for instance, urban synergy is essentially zero (and slightly negative on `land_use` and `junction_type`), while `mismatch_mcq_easy` jumps from 25% with the satellite alone to 66% with both views. We evaluate the raw panel under each model’s native street-view scheme, so it speaks only

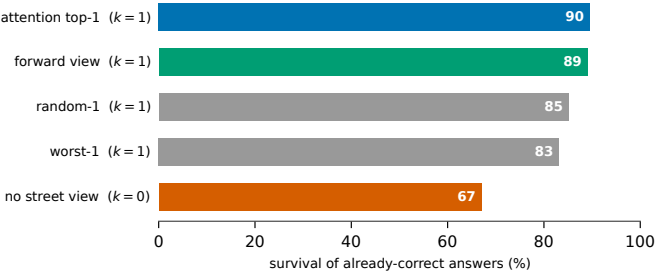


Fig. 5: Survival of already-correct answers as the street-view set is pruned (Qwen3.6-27B). Every bar is the same quantity—the fraction of questions correct with all views that remain correct after pruning. Keeping one street view ($k=1$, any rule) retains 83–90%; keeping none ($k=0$, satellite only) retains 67%. This is retention over already-correct items, never an accuracy.

to whether the qualitative regime split recurs, not to a like-for-like view-count comparison. If anything this grants the panel more street-view evidence to fuse, making the near-zero urban synergy it still shows a conservative reading. The regime split is thus a property of current VLMs and current paired-view tasks, not a fine-tuning artefact.

4.7 Inference-time view pruning

If a single street view carries the urban signal, pruning the others at inference should cost little. We measure this on a stronger raw model (Qwen3.6-27B) with a single metric throughout: among the questions the model already answers correctly with all street views, what fraction *survive*—stay correct—when the street-view set is cut to k images (Fig. 5). At $k=1$, keeping the forward view retains 89% of those correct answers, essentially matching the attention-selected best view (90%) and edging out a random view (85%) or the worst (83%); one well-chosen street view is almost as good as four. Cutting to $k=0$ (no street view, satellite only) still retains 67%—reflecting residual text and label signal—so no *specific* single view is essential, and the gap between informed and random selection is small.

4.8 What do additional street views buy at training time?

Does training with all four street views, rather than one, change the picture? We compare our `satfwd` adapter against an otherwise identical model trained on the marked satellite tile plus all four street views (5-STV), same backbone and seen/unseen split, each evaluated in its own full mode. On the seven urban tasks the 5-STV model averages 67.6% versus `satfwd`’s 65.3%—a 2.3-point gain from training with three additional street views, well below the cost of carrying

four times the visual data. On the cross-view tasks the two are comparable. For urban-attribute deployment, three extra street views buy little.

5 Robustness and Confounds

A templated benchmark and a within-model ablation invite specific confounds—option artefacts, letter skew, image-count asymmetry, shared label priors. We check each, and the headline survives all of them for one structural reason we make precise first: it is a within-item comparison.

The headline is a within-item difference, so shared artefacts cancel. The central result—that full loses to the per-item single-view oracle by 12.3 points—compares two predictions on the *same* question, with byte-identical text and options, differing only in which images are supplied. Any effect that does not depend on the image condition—a label prior, an option-length regularity, a template the model has memorised, a residual text shortcut—contributes equally to both sides and subtracts out of the difference. The interference we measure therefore cannot be manufactured by text leakage: a model that read the answer off the text alone would score identically under **sat**, **sv**, and **full**, yielding zero synergy and zero oracle gap, not the systematic loss we observe—and this holds regardless of how clean the absolute accuracies are. Empirically the model also reads the imagery (Sec. 4.2): each single view scores far above the 25.2% chance rate, so the comparison is between two genuinely vision-driven predictions. No model refused or hedged on any urban item.

Option text and position. Even so, we verify the answer cannot be read off the templated options. On the seven urban-attribute tasks the correct letter is uniform ($A/B/C/D = 662/694/628/645$; $\chi^2 = 3.6$, $p = 0.31$ against uniform; an always-‘A’ baseline scores 25.2%), and the correct option is not systematically the longest—its mean length (25.9 characters) is slightly below that of the distractors (26.5), and it is the single longest option on only 22.9% of items, below the 25% chance rate. Across the full 12-task benchmark the letter distribution is skewed (always-‘A’ 29.3%), an artefact of the option-shuffling RNG; that skew is carried entirely by the cross-view tasks (34.4%) and is absent from the urban tasks evaluated here.

Image count. A concern in multi-view ablations is that one condition may simply present more images than another. In our primary configuration this concern does not arise: the **sat**, **sv**, and **full** conditions supply one, one, and two images respectively, so the comparison of **full** against either single view adds exactly one image of the other modality, with no token-count asymmetry between the two views. (The cross-view control tasks of Sec. 4.3 feed the full street-view set by construction; we treat them as a sensitivity check on the instrument, not as an image-count-matched comparison, and the headline result does not depend on them.)

OSM provenance. Our labels are OSM-derived, and two of the four zero-shot remote-sensing models we compare against—LHRS-Bot-Nova and EarthDial—are trained on OSM-derived text, so their scores may reflect shared OSM priors; we treat them as an upper bound on visual skill and mark them accordingly (Sec. 4.5). This does not touch our main result, which is a *within-model* difference (full vs. the better single view): any view-independent OSM prior cancels in the difference.

Satellite resolution. The redundancy is measured against high-resolution overhead imagery, not a coarse proxy: every benchmark tile resolves to a sub-meter source (ESRI ~ 0.3 m/px to 0.5 m/px, NAIP ~ 0.6 m/px, IGN ~ 0.2 m/px). For tasks that are structurally ground-level no overhead resolution recovers the cue. We cannot rule out that still higher ground sampling distance would help the few tasks where overhead geometry is plausibly resolution-limited (building height, urban density), and flag this as the main open question (Sec. 6.1).

6 Discussion

Complementarity here is a property of the task, not of imagery in general. An attribute question—local land use, building height, transit access—is answerable from one location’s imagery, so the two views carry much the same answer-relevant content and the combined input rarely beats the stronger one. A cross-view question depends instead on the relationship between the views, where our models do use both.

The measurements show the model derives little usable signal from the second view, and that the combined input is on net slightly worse than the better single view; they do not identify the internal cause. Redundant information across views, a representation dominated by one modality, and an architecture that fails to combine them would all produce the pattern we observe, and our data do not distinguish among them. The input–output fact—adding the second view does not help and often hurts—is what a practitioner deciding whether to collect it needs to know.

This has a practical consequence. Street-level and high-resolution satellite imagery differ sharply in cost, coverage, and licensing, yet for these attribute families the second view adds at most a couple of points and degrades about one item in ten. A deployment can serve a single view—the cheaper or better-covered one—at little accuracy cost; and when both are available, the evidence favors *routing* each question to one view over feeding the model both.

6.1 Limitations and scope

The claim concerns seven urban-attribute tasks in four-option MCQ form at commonly served satellite resolutions; it is not a claim that overhead imagery is uninformative or that fusion fails in general, as the cross-view tasks show large synergy on the same models. Three caveats bound it further: our two

training regimes share one held-out test set, of which the held-out-cities split is the true generalization check; higher ground sampling distance might help the few overhead-geometry-limited tasks (building height, urban density); and an open-ended or regression formulation could surface complementary signal an MCQ cannot express.

7 Conclusion

We measured, rather than assumed, the value of a second perspective for urban attribute recognition. Across model scale, training regime, inference-time view count, and prompting strategy, the second view adds at most a couple of points over the better single view, and item by item it removes more correct answers than it adds; the same models do benefit on tasks whose answer lives in the correspondence between the two views. We release OELLM—benchmark, adapters, and protocol—for reuse. The practical lesson is simple: before collecting a second visual modality, ask whether the task structurally requires cross-view reasoning; if it does not, current models will not turn the extra view into accuracy.

8 Future Work

Because only the cross-view tasks consistently benefit from multiple street views, the natural next step is to exploit those views for richer geometric reasoning. The four road-aligned angles per location contain the signal needed for local 3D reconstruction; lifting urban understanding from 2D image fusion to 3D spatial reasoning would let a model reason about occlusion, scale, building geometry, and pedestrian-environment affordances in ways single 2D images cannot support, with OELLM’s four-angle captures as a natural input substrate. Further directions include fusion architectures that combine modalities more effectively on redundant-view tasks, scaling to more cities to reduce the Europe-heavy distribution, and extending deterministic OSM grounding to socio-economic indicators.

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